

# Symbols

This list gives a brief overview of the use of variables in the text. Due to the large number of quantities occurring, the same letter might be used in different chapters for different quantities. Those variables that are used only locally (in a single section), and explained directly at their occurrence, are not mentioned here.

*Lowercase symbols:*

|                       |                                                                   |                            |
|-----------------------|-------------------------------------------------------------------|----------------------------|
| $a_\ell$              | amplitudes of the $\ell$ th MPC                                   | 3, 5, 6, 7, 8, 9           |
| $a_p$                 | auxiliary variables                                               | 4                          |
| $a_{n,m}$             | amplitudes of components from direction $n$ , delay $m$           | 12                         |
| $a_{i,j}$             | expansion functions for video coding                              | 25                         |
| $a(t)$                | exogenous rate                                                    | 20                         |
| $b_m$                 | $m$ th bit                                                        | 10, 11, 12, 13, 14, 18     |
| $cdf$                 | cumulative distribution function                                  |                            |
| $c_i$                 | amplitudes of resolvable MPCs; tap weights for tapped delay lines | 6, 7                       |
| $c_i$                 | complex transmit symbols                                          | 10, 11, 12, 14, 15, 31, 32 |
| $c_0$                 | speed of light                                                    | 4, 5, 12, 15, 29           |
| $d$                   | distance BS–MS                                                    | 4, 29                      |
| $d$                   | distance in signal space diagram                                  | 11, 12, 13                 |
| $d_i$                 | expanded source bits in polar code                                | 13                         |
| $d_{km}$              | Euclidean distance between signal points with index $k$ and $m$   | 11                         |
| $d_{ml}(t)$           | analysis function in generalized OFDM                             | 15                         |
| $d(\vec{x}, \vec{y})$ | distance of codewords                                             | 13                         |
| $d_a$                 | distance between antenna elements                                 | 8, 9, 12                   |
| $d_{\text{break}}$    | distance BS–breakpoint                                            | 3, 4, 7                    |
| $d_{\text{direct}}$   | direct pathlength                                                 | 4                          |
| $d_{\text{div}}$      | diversity order                                                   | 16                         |
| $d_{\text{layer}}$    | thickness of layer                                                | 4                          |
| $d_H$                 | Hamming distance                                                  | 13                         |
| $d_{\text{min}}$      | minimum distance (in a code)                                      | 13                         |
| $d_R$                 | Rayleigh distance                                                 | 4, 8                       |
| $d_{\text{refl}}$     | length of reflected path                                          | 4                          |
| $d_0$                 | distance to reference point                                       | 5, 7                       |
| $e$                   | basis of natural logarithm                                        |                            |
| $e$                   | prediction error                                                  | 24                         |
| $e(t)$                | impulse response of equalizer                                     | 14                         |
| $f$                   | frequency                                                         | 5                          |
| $f_k$                 | impulse response of discrete-time channel                         | 14                         |
| $f_n$                 | subcarrier frequencies in OFDM                                    | 15                         |
| $f(\cdot)$            | function                                                          |                            |
| $f_c$                 | carrier frequency                                                 | 3, 5, 7, 9, 10, 15         |
| $f_{\text{inst}}$     | instantaneous frequency                                           | 5, 11                      |
| $f_{\text{mod}}$      | modulation frequency                                              | 10                         |
| $f_{\text{rep}}$      | repetition frequency                                              | 9                          |
| $f_s$                 | sampling frequency                                                | 17                         |
| $g_m$                 | discrete impulse response of channel plus feedforward filter      | 14                         |
| $\mathbf{g}$          | network encoding vector                                           | 27                         |
| $g(\cdot)$            | function                                                          |                            |
| $g(t)$                | basis pulse                                                       | 10, 11, 14, 15             |
| $\tilde{g}(t)$        | phase pulse                                                       | 10                         |
| $g_N(t)$              | Nyquist raised cosine pulse                                       | 10                         |
| $g_{NR}(t)$           | root Nyquist raised cosine pulse                                  | 10                         |
| $g_R(t)$              | rectangular basis pulse                                           | 10                         |

|                              |                                                     |                                |
|------------------------------|-----------------------------------------------------|--------------------------------|
| $h(t, \tau)$                 | channel impulse response                            | 2, 6, 7, 9, 11, 12, 15, 16, 19 |
| $h(t, \tau, \phi)$           | directionally resolved impulse response             | 7                              |
| $h_{\text{BS}}$              | height of BS                                        | 7                              |
| $h_{\text{UE}}$              | height of UE                                        | 7                              |
| $h_{\text{meas}}(t_i, \tau)$ | measured impulse response                           | 9                              |
| $h_{\text{mod}}$             | modulation index of CPFSK signal                    | 10                             |
| $h_{\text{r,d}}$             | complex channel gain from relay to destination      | 27                             |
| $h_{\text{roof}}$            | height of rooftop                                   | 7                              |
| $h_{\text{r,d}}$             | complex channel gain from relay to destination      | 27                             |
| $h_{\text{RX}}$              | height of RX                                        | 4                              |
| $h_{\text{s}}$               | height of screen                                    | 4                              |
| $h_{\text{s,d}}$             | complex channel gain from source to destination     | 27                             |
| $h_{\text{s,r}}$             | complex channel gain from source to relay           | 27                             |
| $h_{\text{TX}}$              | height of TX                                        | 4                              |
| $\mathbf{h}_{\text{d}}$      | vector of desired impulse responses                 | 12, 16                         |
| $\tilde{h}$                  | Normalized channel impulse response                 | 21                             |
| $i$                          | index counter                                       |                                |
| $j$                          | index counter                                       |                                |
| $j$                          | imaginary unit                                      |                                |
| $k, k_0$                     | wavenumber                                          | 4, 5, 8, 12                    |
| $k$                          | index counter                                       |                                |
| $\mathbf{k}$                 | wave vector                                         | 5                              |
| $k_{\text{B}}$               | Boltzmann constant                                  | 3, 17                          |
| $l$                          | index counter                                       | 8, 15, 16                      |
| $\mathbf{l}$                 | location vector                                     | 8                              |
| $\ell$                       | index counter for MPCs                              |                                |
| $\ell_{\text{eff}}$          | electric length of an antenna                       | 8                              |
| $m$                          | Nakagami $m$ -factor                                | 5, 12                          |
| $m$                          | index counter                                       | 10, 11, 12, 13, 14,            |
| $m$                          | index for parity check bits                         | 13                             |
| $n$                          | index counter                                       |                                |
| $n_n$                        | sampled noise values                                |                                |
| $\mathbf{n}$                 | vector of noise samples                             | 16, 22, 28, 29                 |
| $n_1$                        | refraction index for medium                         | 4                              |
| $n(t)$                       | noise signal                                        | 9, 11, 12, 13, 14, 16, 19, 22  |
| $n_{\text{LP}}(t)$           | low-pass noise                                      | 11                             |
| $n_{\text{BP}}(t)$           | bandpass noise                                      | 11                             |
| $\tilde{n}_m$                | sample values of correlated noise                   | 14                             |
| $p$                          | transition probability; fraction of “good” channels | 13                             |
| $p$                          | parameter in p-persistent MAC                       | 18                             |
| $p(t)$                       | modulated pulse/pulse sequence                      | 9, 10, 15                      |
| $p_{\text{c}}$               | probability of coverage                             | 21                             |
| $p_{\text{r}}$               | probability of being in receive state               | 23                             |
| $p_{\text{t}}$               | probability of being in transmit state              | 23                             |
| $\mathbf{p}$                 | packet vector                                       | 27                             |
| $\mathbf{P}$                 | power allocation vector                             | 20                             |
| $\text{pdf}$                 | probability density function                        |                                |
| $q$                          | Auxiliary variable                                  | 15                             |
| $q_m$                        | impulse response of channel + equalizer             | 14                             |
| $q_m$                        | quantization noise sample                           | 24                             |
| $\mathbf{q}$                 | vector of quantization noise samples                | 22                             |
| $r$                          | absolute value of fieldstrength                     | 5                              |
| $r$                          | distance                                            | 8, 29                          |
| $r$                          | spectral efficiency                                 | 13                             |
| $\mathbf{r}$                 | received signal vector                              | 8, 11, 13                      |
| $\mathbf{r}$                 | position vector                                     | 4, 6, 7, 8, 9, 29              |
| $r(t)$                       | received signal                                     | 8, 13, 14, 23                  |
| $r_{\text{LP}}(t)$           | low-pass representation of received signal          | 11                             |
| $\mathbf{s}$                 | transmit signal vector                              | 8, 9, 13, 16, 22, 31           |
| $s(t)$                       | transmit signal                                     | 8, 9                           |
| $s(\nu, \tau)$               | spreading function                                  | 6                              |
| $s_1(t)$                     | auxiliary signal                                    | 9                              |
| $s_{\text{LP,BP}}(t)$        | low-pass (bandpass) signal                          | 10                             |

|                             |                                              |                       |
|-----------------------------|----------------------------------------------|-----------------------|
| $\mathbf{s}_{\text{LP,BP}}$ | signal vector in low-pass (bandpass)         | 10                    |
| $\mathbf{s}_{\text{synd}}$  | syndrome vector                              | 13                    |
| $t$                         | absolute time                                | 2, 10, 11, 12, 14, 18 |
| $\mathbf{t}$                | precoding vector                             | 16, 22                |
| $\tilde{\mathbf{t}}$        | Best among discrete set of precoding vectors | 16                    |
| $t_0$                       | start time                                   | 6                     |
| $t_s$                       | sampling time                                | 11                    |
| $u$                         | auxiliary variable                           | 10                    |
| $u_i$                       | sequence of sample values at equalizer input | 14                    |
| $u_i$                       | X-coordinate of anchor                       | 29                    |
| $\mathbf{u}$                | vector of information symbols                | 13                    |
| $\mathbf{u}$                | External input for Kalman filter             | 29                    |
| $\mathbf{u}(\phi)$          | steering vector                              | 6, 8                  |
| $v$                         | velocity                                     | 5                     |
| $v_i$                       | y-coordinate of anchor                       | 29                    |
| $\mathbf{v}$                | velocity vector                              | 5                     |
| $\mathbf{v}$                | singular vector                              | 16                    |
| $w_n$                       | antenna weights                              | 8, 12, 16, 22         |
| $w_n$                       | weights for weighted fair queueing           | 20                    |
| $w_n$                       | combiner weights for emitter detection       | 26                    |
| $\mathbf{w}$                | vector of weights                            | 8, 12, 16, 22, 26     |
| $x$                         | x-coordinate                                 | 4, 29                 |
| $x$                         | general variable                             | 5                     |
| $x$                         | transmit signal                              | 13, 26, 27, 28        |
| $x$                         | speech signal                                | 24                    |
| $x(t)$                      | input signal                                 | 6, 17                 |
| $\mathbf{x}$                | code vector, sequence of transmit signals    | 13                    |
| $\mathbf{x}$                | state in Kalman filter                       | 29                    |
| $y$                         | y-coordinate                                 | 4, 29                 |
| $y$                         | received signal                              | 13, 26, 27            |
| $\mathbf{y}$                | sequence of receive signals                  | 13                    |
| $y(t)$                      | output signal                                | 6, 17                 |
| $z$                         | z-coordinate                                 | 8                     |
| $z$                         | Bhattacharyya parameter                      | 13                    |

*Uppercase symbols:*

|                                   |                                                     |                       |
|-----------------------------------|-----------------------------------------------------|-----------------------|
| $A$                               | amplitude of dominant component                     | 5, 7                  |
| $A$                               | state in the trellis diagram                        | 13                    |
| $A$                               | size of an area                                     | 21                    |
| $\mathbf{A}$                      | vector potential                                    | 8.1                   |
| $\mathbf{A}$                      | area                                                | 21                    |
| $\mathbf{A}$                      | auxiliary matrix                                    | 22                    |
| $\mathbf{A}$                      | state transition matrix in Kalman filter            | 29                    |
| $A(d_{\text{TX}}, d_{\text{RX}})$ | amplitude factors for diffraction                   | 4                     |
| $A(\tau, \nu)$                    | ambiguity function                                  | 15                    |
| $A_{\text{RX}}$                   | antenna area of receiver                            | 4                     |
| $ADF$                             | average duration of fades                           |                       |
| $B(\nu, f)$                       | Doppler-variant transfer function coherence         | 6                     |
| $B_{\text{coh}}$                  | bandwidth                                           | 6, 16                 |
| $B$                               | bandwidth                                           | 3, 10, 13, 16, 17, 29 |
| $B$                               | state in the trellis diagram                        | 13                    |
| $\mathbf{B}$                      | Auxiliary matrix                                    | 29                    |
| $B_{\text{coh}}$                  | coherence bandwidth                                 | 6                     |
| $B_G$                             | bandwidth of Gaussian filter                        | 10                    |
| $B_n$                             | noise bandwidth                                     | 11                    |
| $B_r$                             | receiver bandwidth                                  | 11                    |
| $B_{\text{stat}}$                 | stationarity bandwidth                              | 7                     |
| $BER$                             | bit error probability                               | 11                    |
| $C$                               | capacity (often used loosely for "achievable rate") | 13, 16, 18, 21, 22    |
| $C$                               | proportionality constant                            |                       |
| $C$                               | state in the trellis diagram                        | 13                    |

|                                 |                                                                                |                        |
|---------------------------------|--------------------------------------------------------------------------------|------------------------|
| $C(\mathbf{x})$                 | cost function                                                                  | 29                     |
| $\mathbf{C}$                    | auxiliary matrix                                                               | 22, 29                 |
| $\mathbf{C}$                    | covariance matrix                                                              |                        |
| $\mathbf{C}$                    | space-time coding matrix                                                       | 16                     |
| $\mathbf{C}_p$                  | Pilot sequence                                                                 | 16                     |
| $C_{\text{crest}}$              | crest factor                                                                   | 9                      |
| $C_{\text{CSIT}}$               | capacity with channel state information at the TX                              | 16, 22                 |
| $CW$                            | contention window                                                              | 18, 33                 |
| $D$                             | diffraction coefficient                                                        | 4                      |
| $D$                             | distance between BSs                                                           | 3, 21                  |
| $D$                             | quadratic form                                                                 | 11                     |
| $D$                             | state in the trellis diagram                                                   | 13                     |
| $D$                             | unit delay                                                                     |                        |
| $D$                             | radius of sphere in sphere decoder                                             | 16                     |
| $D$                             | maximum distortion                                                             | 24                     |
| $D()$                           | antenna directivity                                                            | 8                      |
| $\tilde{D}_{N_t}$               | distance measure in sphere decoder                                             | 16                     |
| $D_{\text{leav}}$               | interleaver separation                                                         | 13                     |
| $E$                             | electric fieldstrength                                                         | 4                      |
| $E(f)$                          | transfer function of equalizer                                                 | 14                     |
| $\mathbf{E}$                    | error matrix in space-time code                                                | 16                     |
| $\mathbf{E}$                    | MSE matrix                                                                     | 22                     |
| $E_{\text{diff}}$               | fieldstrength of diffracted field                                              | 4                      |
| $E_{\text{inc}}$                | fieldstrength of incident field                                                | 4                      |
| $E\{\}$                         | expectation                                                                    | 4, 12, 13, 19          |
| $E_0$                           | normalization fieldstrength                                                    | 12                     |
| $E_B$                           | bit energy                                                                     | 10, 11, 12, 13         |
| $E_c$                           | energy of coded bit                                                            | 13                     |
| $E_C$                           | chip energy                                                                    | 19                     |
| $E_S$                           | symbol energy                                                                  | 10, 11, 12             |
| $E_{s,k}$                       | energy of $k$ th signal                                                        | 10, 19                 |
| $E1, E2$                        | fieldstrength of multipath components                                          | 5                      |
| $F$                             | noise figure                                                                   | 3, 17                  |
| $F$                             | local mean of fieldstrength                                                    | 5                      |
| $\mathbf{F}$                    | encoding matrix for polar code                                                 | 13                     |
| $\mathbf{F}$                    | auxiliary matrix                                                               | 20, 22                 |
| $F(\nu_F)$                      | fresnel integral                                                               | 4                      |
| $F(z)$                          | factorization of the transfer function of the equivalent time discrete channel | 14                     |
| $F_q$                           | finite field of size $q$                                                       | 27                     |
| $\tilde{F}$                     | modified Fresnel integral                                                      | 4                      |
| $FM$                            | Fading margin                                                                  | 3, 5                   |
| $\mathcal{F}$                   | Fourier transform                                                              |                        |
| $G$                             | gain of component                                                              | 17                     |
| $G$                             | gain of an amplifier stage                                                     | 3, 17                  |
| $G(D)$                          | code polynomial                                                                | 31, 32, 33             |
| $G(\gamma), G(\varphi, \theta)$ | antenna pattern                                                                | 5, 8                   |
| $G(\nu, \nu_1, \nu_2)$          | Gaussian function                                                              | 7                      |
| $G(v)$                          | probability generating functional                                              | 21                     |
| $\mathbf{G}$                    | generator matrix                                                               | 13                     |
| $\mathbf{G}$                    | global encoding matrix                                                         | 27                     |
| $G_{\text{code}}$               | code gain                                                                      | 13                     |
| $G_e$                           | Antenna element gain                                                           | 16                     |
| $G_{\text{max}}$                | maximum gain                                                                   |                        |
| $G_N$                           | spectrum of Nyquist pulse                                                      | 10                     |
| $G_{NR}$                        | spectrum of root Nyquist pulse                                                 | 10                     |
| $G_R$                           | spectrum of rectangular pulse                                                  | 10                     |
| $G_{RX}$                        | antenna power gain of receive antenna                                          | 3, 4, 8, 9, 16, 22, 23 |
| $G_T$                           | transducer gain                                                                | 17                     |
| $G_{TX}$                        | antenna power gain transmit antenna                                            | 3, 4, 8, 9, 16, 22, 23 |
| $\mathbf{G}_G$                  | matrix with iid Gaussian entries                                               | 7, 16                  |
| $\tilde{G}$                     | Complex antenna gain                                                           | 3, 8, 9, 16            |

|                           |                                                                              |                        |
|---------------------------|------------------------------------------------------------------------------|------------------------|
| $H$                       | transfer function of the channel                                             | 5, 6, 7, 9, 15, 16, 22 |
| $H(X)$                    | entropy                                                                      | 13, 25                 |
| $\mathbf{H}$              | parity check matrix                                                          | 13                     |
| $\mathbf{H}$              | matrix of transfer functions                                                 | 6, 7, 9, 16, 22, 33    |
| $\mathbf{H}$              | Hessian matrix                                                               | 29                     |
| $H_b(p)$                  | binary entropy function                                                      | 16                     |
| $H_D(X)$                  | entropy of binary symmetric channel                                          | 13                     |
| $H_R(f)$                  | transfer function of receive filter                                          | 11, 19                 |
| $\mathbf{H}_{\text{had}}$ | Hadamard matrix                                                              | 15, 19                 |
| $\tilde{\mathbf{H}}$      | matrix of normalized transfer functions                                      | 16, 22                 |
| $\hat{\mathbf{H}}$        | estimated channel matrix                                                     | 16, 22                 |
| $\bar{\mathbf{H}}$        | interference channel matrix                                                  | 22                     |
| $\mathbf{H}^+$            | pseudo-inverse of $\mathbf{H}$                                               |                        |
| $\mathcal{H}_0$           | null hypothesis                                                              | 26                     |
| $I$                       | number of time symbols in GFDM or OTFS                                       | 15                     |
| $I$                       | total interference power                                                     | 21                     |
| $I$                       | mutual information                                                           | 27                     |
| $I(t)$                    | in-phase component                                                           | 5                      |
| $I(t)$                    | link control action                                                          | 20, 23                 |
| $I(x, y)$                 | mutual information                                                           | 13, 16                 |
| $I_0$                     | modified Bessel function                                                     | 5, 11                  |
| $\mathbf{I}$              | identity matrix                                                              | 15, 16, 22             |
| $\mathbf{I}(\mathbf{x})$  | Fisher information matrix                                                    | 29                     |
| $\mathcal{I}$             | set of codeword indices that carry information                               | 13                     |
| $INR$                     | interference-to-noise ratio                                                  | 23                     |
| $ J $                     | Jacobi determinant                                                           | 5                      |
| $J_0$                     | bessel function zero order first kind                                        |                        |
| $J_k$                     | rank of interference channel matrix                                          | 22                     |
| $\mathbf{J}_k$            | auxiliary matrix for precoder optimization                                   | 22                     |
| $K$                       | scaling constant for STDCC                                                   | 8                      |
| $K$                       | number of resolvable directions                                              | 9                      |
| $K$                       | number of bits in a symbol                                                   | 10                     |
| $K$                       | number of information symbols in a codeword                                  | 13                     |
| $K$                       | number of equalizer taps ( $2K + 1$ )                                        | 14                     |
| $K$                       | number of users                                                              | 18, 19, 20, 21, 22, 23 |
| $K$                       | number of relays                                                             | 27                     |
| $\mathbf{K}$              | Kalman gain                                                                  | 29                     |
| $\hat{K}$                 | total number of users in the cell (even when not active)                     | 22                     |
| $K_r$                     | rice factor                                                                  | 5                      |
| $K(t, \tau)$              | kernel function                                                              | 6                      |
| $K_I$                     | system margin                                                                | 3                      |
| $L$                       | diffraction loss                                                             | 4                      |
| $L$                       | number of cells in convolutional encoder                                     | 13                     |
| $L$                       | duration of the impulse response of the equivalent time discrete channel $f$ | 14                     |
| $L$                       | determinant of Gabor basis                                                   | 15                     |
| $L$                       | number of cells                                                              | 21, 22                 |
| $L$                       | number of RF chains in HS-MRC                                                | 16, 22                 |
| $L_i$                     | attenuation at the $i$ th screen                                             | 4                      |
| $L, L(u)$                 | log-likelihood ratio                                                         | 13                     |
| $L(..)$                   | Lagrangian                                                                   | 20                     |
| $L(\mathbf{Q}(t))$        | Lyapunov function                                                            | 20, 23                 |
| $\mathbf{L}$              | coefficient matrix                                                           | 27                     |
| $L_a$                     | antenna dimension                                                            | 4                      |
| $L_c$                     | correlation length                                                           | 4                      |
| $L_c$                     | channel reliability                                                          | 13                     |
| $L_c$                     | length of channel impulse response                                           | 14                     |
| $L_c$                     | length of space-time-code codeword                                           | 16                     |
| $L_e$                     | extrinsic log-likelihood                                                     | 13                     |
| $L_f$                     | losses in feeder                                                             | 3                      |
| $L_{\text{msd}}$          | multiscreenloss                                                              | 7                      |
| $L_{\text{ori}}$          | street orientation loss                                                      | 7                      |

|                                  |                                                           |                   |
|----------------------------------|-----------------------------------------------------------|-------------------|
| $L_{\text{rts}}$                 | diffraction loss rooftop to street                        | 7                 |
| $L_{\text{symb}}$                | number of symbols where two possible sequences differ     | 13                |
| $L_{\text{Tr}}$                  | truncation depth                                          | 13                |
| $L_{\infty}(\mathbf{H})$         | capacity loss at high SNR                                 | 16, 22            |
| $\tilde{L}$                      | dimension of space-time code                              | 16                |
| $M$                              | number of elements in the alphabet                        | 10, 13, 15        |
| $M$                              | maximum of number of transmit, receive antennas           | 16                |
| $M$                              | number of clusters                                        | 7                 |
| $M$                              | dimension of the signal space                             | 28                |
| $M(s)$                           | moment-generating function                                | 11, 12            |
| $M(\phi, \theta)$                | array factor                                              | 8, 16             |
| $M_p$                            | length of pilot sequence                                  | 16, 22            |
| $N$                              | number of screens                                         | 4                 |
| $N$                              | number of MPCs                                            | 5, 7, 9, 15       |
| $N$                              | size of the set of expansion functions                    | 11                |
| $N$                              | total number of symbols in the code                       | 13                |
| $N$                              | number of mod-2 adders in the convolutional encoder       | 13                |
| $N$                              | number of received symbols in Viterbi equalizer           | 14                |
| $N$                              | number of subcarriers                                     | 15                |
| $N$                              | min of number of transmit, receive antennas               | 16                |
| $N$                              | length of input block for video coding                    | 25                |
| $N$                              | number of samples used for detecting emitter              | 26                |
| $N(f)$                           | noise spectrum                                            | 6, 11             |
| $\tilde{N}$                      | number of taps in a tapped delay line                     | 7, 15             |
| $\tilde{N}$                      | number of bits for convolutional encoder in TCM noise     | 13                |
| $N_0$                            | power-spectral density                                    | 3, 11, 17         |
| $N_{\text{BS}}$                  | number of BS antennas                                     | 16, 22            |
| $N_{\text{C}}$                   | number of channels                                        | 20                |
| $N_{\text{cluster}}$             | cluster size                                              | 21                |
| $N_{\text{f}}$                   | pilot spacing in frequency                                | 15                |
| $N_{\text{IO}}$                  | number of interacting objects                             | 16                |
| $N_{\text{r}}$                   | number of receive antennas                                | 9, 12, 16, 22     |
| $N_{\text{r}}$                   | number of ranging symbols                                 | 29                |
| $N_{\text{R}}$                   | level crossing rate                                       | 5                 |
| $N_{\text{reg}}$                 | length of shift register                                  | 13                |
| $N_{\text{RF}}$                  | number of RF chains                                       | 33                |
| $\mathbf{N}_{\text{subchannel}}$ | number of spatial streams                                 | 16, 33            |
|                                  | number of subchannels                                     | 13                |
| $N_{\text{symb}}$                | number of information bits/symbols in TCM                 | 15                |
| $N_{\text{t}}$                   | pilot spacing in time                                     | 16, 22            |
| $N_{\text{t}}$                   | number of transmit antennas                               | 16, 22            |
| $N_{\text{UE}}$                  | number of UE antennas                                     | 5, 13, 15, 16, 18 |
| $P$                              | power                                                     | 7                 |
| $P(t, \tau)$                     | instantaneous PDP                                         | 13                |
| $P(\phi, \phi')$                 | probability of transitioning from state $\phi$ to $\phi'$ | 20, 21, 22        |
| $P_k$                            | power of the $k$ th user                                  | 13                |
| $\mathbf{P}$                     | non-systematic part of block encoder matrix spreading     | 15                |
| $\mathbf{P}$                     | matrix for multicarrier-CDMA                              | 16, 22            |
| $\mathbf{P}$                     | power allocation matrix                                   | 29                |
| $\mathbf{P}$                     | error covariance matrix                                   | 5                 |
| $\bar{P}$                        | average power                                             |                   |
| $P_0$                            | Total power in relay setup                                | 27                |
| $P_{\text{1dB}}$                 | power at 1 dB compression point                           | 17                |
| $P_{\text{f}}$                   | false alarm probability                                   | 26                |
| $P_{h,S,B}$                      | cross-power spectral densities                            | 6                 |
| $P_{\text{h}}(\tau)$             | PDP                                                       | 6                 |
| $P_{\text{inst}}$                | instantaneous received power                              | 11                |
| $P_{\text{m}}$                   | average power                                             | 5                 |
| $P_{\text{max}}$                 | maximum TX power                                          | 16, 22            |
| $P_{\text{md}}$                  | missed detection probability                              | 26                |
| $P_{\text{n}}$                   | noise power                                               | 3, 17, 27, 28     |
| $P_{\text{pair}}$                | pairwise error probability                                | 11                |

|                         |                                                       |                                |
|-------------------------|-------------------------------------------------------|--------------------------------|
| $P_r$                   | transmit power of relay                               | 27                             |
| $P_{RX}$                | receive power                                         |                                |
| $P_s$                   | signal power                                          | 3, 10                          |
| $P_s$                   | transmit power of source                              | 27                             |
| $P_{TX}$                | transmit power                                        |                                |
| $Pr$                    | probability                                           |                                |
| $Pr_{out}$              | outage probability                                    | 5, 13, 27                      |
| $PG$                    | path gain                                             | 7                              |
| $\overline{PG}$         | average path gain                                     | 7                              |
| $PL$                    | pathloss                                              | 4, 5, 7                        |
| $\overline{PL}$         | average pathloss                                      | 7                              |
| $Q$                     | antenna quality                                       | 8                              |
| $Q$                     | codebook size                                         | 16                             |
| $Q$                     | number of bits in a quantizer                         | 17                             |
| $Q$                     | quantization function                                 | 25                             |
| $Q, Q(t)$               | queue backlog                                         | 20, 23                         |
| $Q(x)$                  | Q-function                                            | 5, 11, 12, 13, 21, 29          |
| $Q(t)$                  | quadrature component                                  | 5                              |
| $Q(z)$                  | transfer function of equivalent channel and equalizer | 14                             |
| $Q(z)$                  | quantization noise                                    | 24                             |
| $Q_M$                   | Marcum's Q-function                                   | 11                             |
| $Q_T$                   | interference quotient                                 | 6                              |
| $R$                     | radius of circle                                      | 5                              |
| $R$                     | cell size                                             | 3, 21                          |
| $R$                     | transmission rate                                     | 13, 18, 20, 22, 23, 26, 27, 28 |
| $R$                     | resistance                                            | 17                             |
| $\mathbf{R}$            | vector of rates                                       | 20                             |
| $\mathbf{R}$            | correlation matrix                                    |                                |
| $\overline{R}$          | average data rate                                     | 20                             |
| $R_B$                   | bit rate                                              | 10                             |
| $R_c$                   | code rate                                             | 13                             |
| $R_e$                   | rank of error matrix                                  | 16                             |
| $R_h$                   | impulse response correlation function                 | 6                              |
| $R_H$                   | frequency correlation function                        | 6                              |
| $R_H$                   | rank of $H$ matrix                                    | 16                             |
| $R_{rad}$               | radiation resistance                                  | 8                              |
| $R_S$                   | symbol rate                                           | 10                             |
| $R_{th}$                | threshold transmission rate                           | 27                             |
| $R_x(i, j)$             | $i, j$ th entry of covariance matrix                  | 25                             |
| $R_{xx}$                | autocorrelation function of $x$                       | 10                             |
| $\tilde{R}_{yy}(t, t')$ | autocovariance signal of received signal              | 6                              |
| $\mathbf{R}_{hh}$       | correlation matrix of channels on subcarriers         | 15                             |
| $\mathbf{R}_{ni}$       | noise and interference correlation matrix             | 16                             |
| $\mathbf{R}_{RX}$       | receive channel correlation matrix                    | 6, 7, 16                       |
| $\mathbf{R}_{TX}$       | transmit channel correlation matrix                   | 6, 7, 16                       |
| $\mathbf{R}_{xx}$       | correlation matrix of $x$                             | 9, 16                          |
| $S$                     | shadowing                                             | 7, 29                          |
| $S_{11}$                | S-parameter                                           | 17                             |
| $S(f)$                  | power spectrum                                        | 5, 6, 10, 11                   |
| $S(t)$                  | topology state                                        | 21, 23                         |
| $S(\nu, \tau)$          | spreading function                                    | 6                              |
| $\mathbf{S}$            | S-parameter matrix                                    | 17                             |
| $S_D(\nu, \tau)$        | Doppler spectrum                                      | 7                              |
| $S_{LP,BP}(f)$          | power spectrum of LP (BP) signal                      | 10                             |
| $S_N$                   | noise power-spectral density                          |                                |
| $S_\tau$                | delay spread                                          | 6, 7, 11, 12, 15               |
| $S_\phi$                | angular spread                                        | 6, 7, 12                       |
| $S_\nu$                 | Doppler spread                                        | 6, 11, 12, 15                  |
| $SER$                   | symbol error probability                              | 11                             |
| $SF$                    | spreading factor                                      | 19, 21, 22                     |
| $SIR$                   | signal-to-interference ratio                          | 5                              |

|                         |                                                         |                |
|-------------------------|---------------------------------------------------------|----------------|
| $SINR$                  | signal-to-interference-and-noise ratio                  | 16, 22         |
| $T$                     | transmission coefficient                                | 4              |
| $T$                     | duration (general)                                      |                |
| $\mathbf{T}$            | auxiliary matrix                                        | 9              |
| $\mathbf{T}$            | measured throughput                                     | 20             |
| $\mathbf{T}$            | transmit beamforming matrix                             | 16, 22         |
| $T_0$                   | reference temperature                                   | 3, 17          |
| $T_B$                   | bit duration                                            |                |
| $T_C$                   | chip duration                                           | 19, 29         |
| $T_{\text{coh}}$        | coherence time                                          | 6              |
| $T_{\text{cp}}$         | duration of cyclic prefix                               | 15             |
| $T_d$                   | delay of pulse in PPM                                   | 10             |
| $T_e$                   | temperature of environment                              | 3, 17          |
| $T_f$                   | frame duration                                          | 19, 29         |
| $T_g$                   | group delay                                             | 11             |
| $T_m$                   | mean delay                                              | 6              |
| $T_m(t)$                | instantaneous mean delay                                | 6              |
| $T_p$                   | packet duration                                         | 18             |
| $T_{\text{per}}$        | periodicity                                             | 10             |
| $T_{\text{rep}}$        | repetition time of pulse signal                         | 9              |
| $T_s$                   | sampling time                                           | 10             |
| $T_S$                   | symbol duration                                         |                |
| $T_{\text{tr}}$         | offered traffic                                         | 20             |
| $\mathcal{T}_0$         | set of states corresponding to transmitted 0            | 13             |
| $\hat{\mathbf{T}}$      | normalized beamforming matrix ( <i>not</i> an estimate) | 16, 22         |
| $TB$                    | time bandwidth product                                  | 9              |
| $U$                     | utility function                                        | 20             |
| $\mathbf{U}$            | unitary matrix                                          | 8, 15, 16, 22  |
| $V$                     | trade-off factor in drift-plus-penalty                  | 20, 23         |
| $V$                     | voltage                                                 | 17             |
| $\mathbf{V}$            | unitary matrix                                          | 8, 16, 22      |
| $W$                     | system bandwidth                                        | 6, 15, 19      |
| $W(\Delta_r)$           | spatial correlation function                            | 4              |
| $W(z)$                  | filter function                                         | 24             |
| $\mathbf{W}$            | receiver filter matrix                                  | 16, 22, 28     |
| $W_a$                   | delay window                                            | 6              |
| $X$                     | complex Gaussian random variable                        | 11             |
| $X(x)$                  | code polynomial                                         | 13             |
| $X^{(u)}(k)$            | Zadoff-Chu sequence                                     | 31, 32         |
| $Y$                     | complex Gaussian random variable                        | 11             |
| $Z$                     | complex Gaussian random variable                        | 11             |
| $Z(t)$                  | virtual queue backlog                                   | 21, 23         |
| <i>Lowercase Greek:</i> |                                                         |                |
| $\alpha$                | complex channel gain                                    | 11, 13         |
| $\alpha$                | rolloff factor                                          | 10             |
| $\alpha$                | fairness coefficient                                    | 20             |
| $\alpha_k$              | fraction of TX power assigned to $k$ th user            | 28             |
| $\alpha_n^2$            | channel power gain on the $n$ th subcarrier             | 15             |
| $\alpha_{\text{diel}}$  | dielectric length                                       | 4              |
| $\alpha_{\text{PL}}$    | pathloss coefficient                                    | 3, 7           |
| $\alpha_{\text{PL},2}$  | pathloss coefficient beyond breakpoint                  | 3, 7           |
| $\beta$                 | decay time constant                                     | 7              |
| $\beta$                 | amplification at relay                                  | 27             |
| $\beta$                 | combined pathloss and shadowing                         | 16, 22         |
| $\beta$                 | fraction of allocated time-frequency resources          | 20             |
| $\beta$                 | effective bandwidth                                     | 29             |
| $\beta_{\text{PL}}$     | pathloss fit offset                                     | 7              |
| $\gamma$                | SNR                                                     | 13, 20         |
| $\gamma$                | angle for Doppler shift                                 | 5, 6           |
| $\bar{\gamma}$          | mean SNR                                                |                |
| $\gamma_B$              | $E_B/N_0$                                               | 11, 12, 14, 15 |



|                            |                                            |                |
|----------------------------|--------------------------------------------|----------------|
| $\gamma_{\text{EGC}}$      | SNR at output of equal gain combiner       | 13             |
| $\gamma_{\text{Euler}}$    | Euler's constant                           | 16             |
| $\gamma_{\text{MRC}}$      | SNR at output of maximum ratio combiner    | 12             |
| $\gamma_{\text{S}}$        | $E_S/N_0$                                  | 11             |
| $\gamma_{\text{th,op}}$    | SNR threshold for operation                | 3              |
| $\gamma_{\text{TX}}$       | SNR at the transmitter                     | 16, 22         |
| $\tilde{\gamma}_k$         | threshold/target SINR for the $k$ th user  | 20, 22         |
| $\delta$                   | complex dielectric constant                | 4              |
| $\delta_{ik}$              | Kronecker delta                            | 12, 14, 15     |
| $\delta(\tau)$             | Dirac function                             | 11, 12, 14, 19 |
| $\varepsilon$              | dielectric constant                        | 4              |
| $\varepsilon$              | transmit power control coefficient         | 21             |
| $\varepsilon$              | distance from rate region boundary         | 20, 23         |
| $\varepsilon$              | relative timing error                      | 29             |
| $\mathbf{e}$               | error vector of a code symbol              | 13             |
| $\mathbf{e}$               | error vector of Kalman filter              | 29             |
| $\varepsilon_i$            | error signal                               | 14             |
| $\varepsilon_k$            | channel estimation errors                  | 16             |
| $\varepsilon_{\text{eff}}$ | effective relative dielectric constant     | 4              |
| $\varepsilon_{\text{f}}$   | frequency offset                           | 15             |
| $\varepsilon_{\text{r}}$   | relative dielectric constant               | 4, 29          |
| $\varepsilon_{\text{WF}}$  | normalization constant for waterfilling    | 15, 16         |
| $\varphi$                  | orientation of a street                    | 7              |
| $\varphi$                  | phase of an MPC                            | 3, 5           |
| $\varphi_m(t)$             | base functions for expansion               | 10             |
| $\tilde{\varphi}$          | deterministic phaseshift                   | 7              |
| $\phi$                     | azimuth angle                              | 4, 6, 7        |
| $\phi$                     | realization of a Trellis state             | 13             |
| $\phi_i$                   | DOA of $i$ th wave                         | 12             |
| $\phi_0$                   | nominal DOA                                | 7, 12          |
| $\phi_{\text{RX}}$         | angle RX - wedge                           | 4              |
| $\phi_{\text{TX}}$         | angle TX - wedge                           | 4              |
| $\eta(t)$                  | $g(t) * h(t)$                              | 14             |
| $\eta_{\text{rad}}$        | radiation efficiency                       | 8              |
| $\kappa$                   | auxiliary variable                         | 12, 15         |
| $\lambda, \lambda_0$       | wavelength                                 | 3, 4, 8, 9, 16 |
| $\lambda$                  | density of a Poisson process               | 7, 21          |
| $\lambda$                  | Lagrange multiplier                        | 20             |
| $\lambda_{i,j}$            | belief of variable node                    | 13             |
| $\lambda(x)$               | auxiliary function in computation of LLR   | 13             |
| $\lambda_{\text{p}}$       | packet transmission rate                   | 18             |
| $\lambda_i$                | $i$ th eigenvalue                          | 16, 22         |
| $\mu$                      | metric                                     | 11             |
| $\mu$                      | stepwidth of LMS                           | 14             |
| $\mu_{i,j}$                | belief of constraint node                  | 13             |
| $\mu(t)$                   | transmission matrix                        | 21, 23         |
| $\mu(x, y)$                | auxiliary function in computation of LLR   | 13             |
| $\nu$                      | Doppler shift                              | 6              |
| $\nu(x)$                   | auxiliary function in computation of LLR   | 13             |
| $\nu_0$                    | subcarrier spacing in generalized OFDM     | 15             |
| $\nu_k$                    | average exogenous rate                     | 20             |
| $\nu_{\text{F}}$           | Fresnel parameter                          | 5              |
| $\nu_{\text{max}}$         | maximum Doppler shift                      | 5, 6, 7, 9     |
| $\nu_{\text{m}}$           | mean Doppler shift                         | 5              |
| $\pi$                      | 3.14                                       |                |
| $\pi(k)$                   | permutation                                | 22             |
| $\pi_{\mathbf{h}}$         | probability for channel state $\mathbf{h}$ | 20             |
| $\omega$                   | angular frequency                          | 4, 8, 17       |
| $\rho$                     | position vector                            |                |
| $\rho$                     | reflection coefficient                     | 4, 16          |
| $\rho_{km}$                | correlation coefficients between signals   | 10             |

|                         |                                                             |               |
|-------------------------|-------------------------------------------------------------|---------------|
| $\sigma$                | standard deviation                                          | 5             |
| $\sigma_i$              | $i$ th singular value                                       | 16,22         |
| $\sigma_c$              | conductivity                                                | 4             |
| $\sigma_e^2$            | variance of the channel estimation error                    | 16            |
| $\sigma_F$              | standard deviation of local mean                            | 5             |
| $\sigma_G$              | standard deviation of Gaussian pulse                        | 10            |
| $\sigma_h$              | standard deviation of height                                | 4             |
| $\sigma_n^2$            | noise variance/power (used synonymous with $P_n$ )          |               |
| $\sigma_S^2$            | power in symbol sequence                                    | 10            |
| $\tau$                  | delay                                                       | 22            |
| $\tau_0$                | symbol spacing in generalized OFDM                          | 15            |
| $\tau_i$                | delay of the $i$ th MPC                                     | 7             |
| $\tau_k$                | fraction of time assigned to $k$ th user                    | 7             |
| $\tau_{Gr}$             | group delay                                                 | 5, 11         |
| $\tau_{max}$            | maximum excess delay                                        | 6, 9, 15      |
| $\theta$                | elevation                                                   | 4, 6, 7, 8, 9 |
| $\theta$                | threshold                                                   | 21, 26        |
| $\theta$                | timing offset                                               | 15            |
| $\boldsymbol{\theta}$   | parameter vector                                            | 9, 29         |
| $\chi_i(t)$             | distortion of the $i$ th MPC                                | 6             |
| $\zeta$                 | ACF of $\eta(t)$                                            | 14            |
| $\xi$                   | SNR loss for discrete precoding                             | 16            |
| $\bar{\xi}$             | penalty function                                            | 23            |
| $\xi_n(t)$              | noise correlation function                                  | 11            |
| $\xi_s(t)$              | FT of the normalized Doppler spectrum                       | 11            |
| $\xi_s(\nu, \tau)$      | scattering function                                         | 11            |
| $\xi_h(t, \tau)$        | FT of the scattering function                               | 11            |
| $\psi$                  | auxiliary angle                                             | 4, 11         |
| $\psi$                  | angle of incidence $90 - \Theta_e$                          | 4             |
| $\psi$                  | phase of channel fading                                     | 5             |
| $\psi_i$                | sample values at output of matched filter                   | 14            |
| $\psi(x)$               | di-gamma function                                           | 16            |
| $\chi$                  | transmit alphabet                                           | 13            |
| $\chi$                  | Threshold for ranging algorithm                             | 29            |
| $\chi_k^2$              | chi-square distributed variable with $k$ degrees of freedom | 16            |
| <i>Uppercase Greek:</i> |                                                             |               |
| $\Delta h_b$            | $= h_b - h_{roof}$                                          | 7             |
| $\Delta x_s$            | distance between measurement points                         | 9             |
| $\Delta \tau_{min}$     | minimum resolvable $\tau$                                   | 9             |
| $\Delta f_{chip}$       | difference in chip frequency                                | 9             |
| $\Delta \varphi$        | angle difference of paths                                   | 4             |
| $\Delta \tau$           | runtime difference                                          | 5             |
| $\Delta \nu$            | Doppler shift                                               | 5             |
| $\Delta$                | phaseshift between two antenna elements                     | 8             |
| $\Delta$                | Lyapunov drift                                              | 21, 23        |
| $\Delta_C$              | determinant of $\mathbf{C}$                                 | 12            |
| $\Delta \phi$           | angular range                                               | 12            |
| $\Phi$                  | Trellis state                                               | 13            |
| $\Phi$                  | Poisson point process                                       | 21            |
| $\Phi$                  | counting measure for Poisson point process                  | 21            |
| $\Phi$                  | pilot tone matrix                                           | 22            |
| $\Phi_{CPFSK}(t)$       | phase of transmit signal for CPFSK signal                   | 10            |
| $\Phi_H$                | phase of the channel transfer function                      | 5, 11         |
| $\Phi_{TX}(t)$          | phase of transmit signal                                    |               |
| $\Lambda$               | lattice                                                     | 15            |
| $\Lambda$               | matrix of eigenvalues                                       | 16            |
| $\Gamma$                | reflection coefficient at circuit transition                | 17            |
| $\Gamma$                | threshold power                                             | 23            |
| $\Gamma(x)$             | Euler's gamma function                                      | 21            |

|                      |                                            |        |
|----------------------|--------------------------------------------|--------|
| $\Omega$             | direction of departure – spatial angle     | 3, 6   |
| $\Omega_n$           | $n$ th moment of Doppler spectrum          | 5, 12  |
| $\Theta(t)$          | queue backlog                              | 21, 23 |
| $\Theta(f(N))$       | Order of (big-O notation)                  | 23.8   |
| $\Theta_e$           | angle of incidence                         | 4      |
| $\Theta_r$           | angle of reflection                        | 4      |
| $\Theta_t$           | angle of transmission                      | 4      |
| $\Theta_n$           | transmission phase of $n$ th bit           |        |
| $\Theta_d$           | diffraction angle                          |        |
| $\Psi$               | direction of arrival – spatial angle       | 3, 6   |
| $\Sigma$             | diagonal matrix containing singular values | 16, 22 |
| $\Xi$                | Fourier transform of $\zeta_i$             | 14     |
| <i>Symbols</i>       |                                            |        |
| $\bar{x}$            | $= E\{x\}$                                 |        |
| $\dot{r}$            | $= dr/dt$                                  |        |
| $\hat{x}$            | estimate of $x$                            |        |
| $x^*$                | complex conjugate                          | $x^*$  |
| $\mathbf{U}^\dagger$ | Hermitian transpose                        |        |
| $\mathbf{U}^T$       | transpose                                  |        |